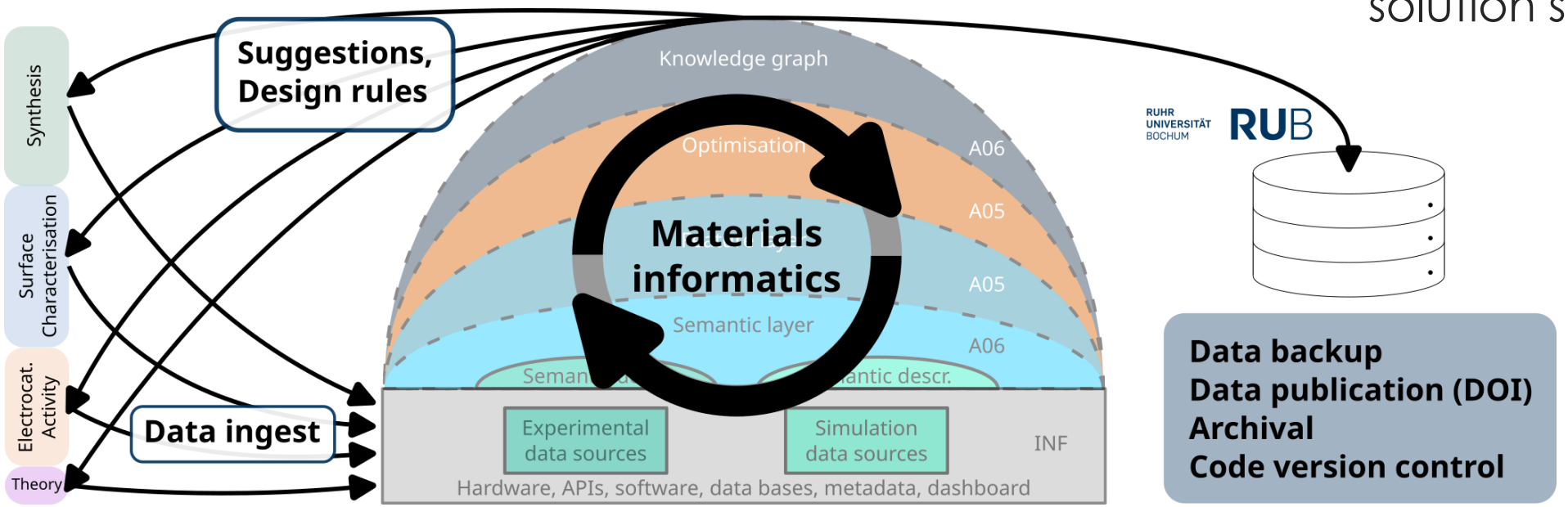
Atomic-scale understanding and design of multifunctional compositionally complex solid solution surfaces

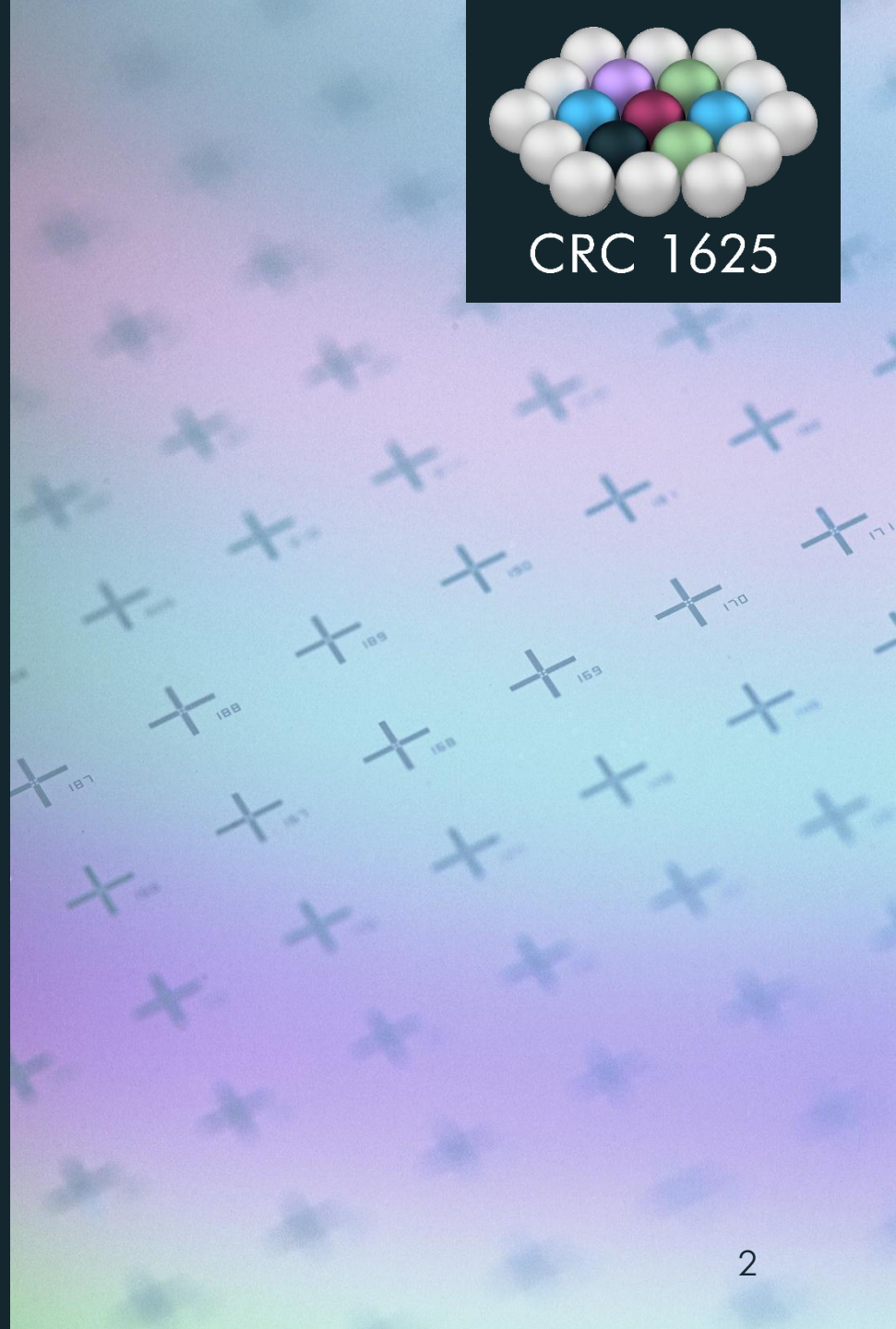
CRC 1625



INF

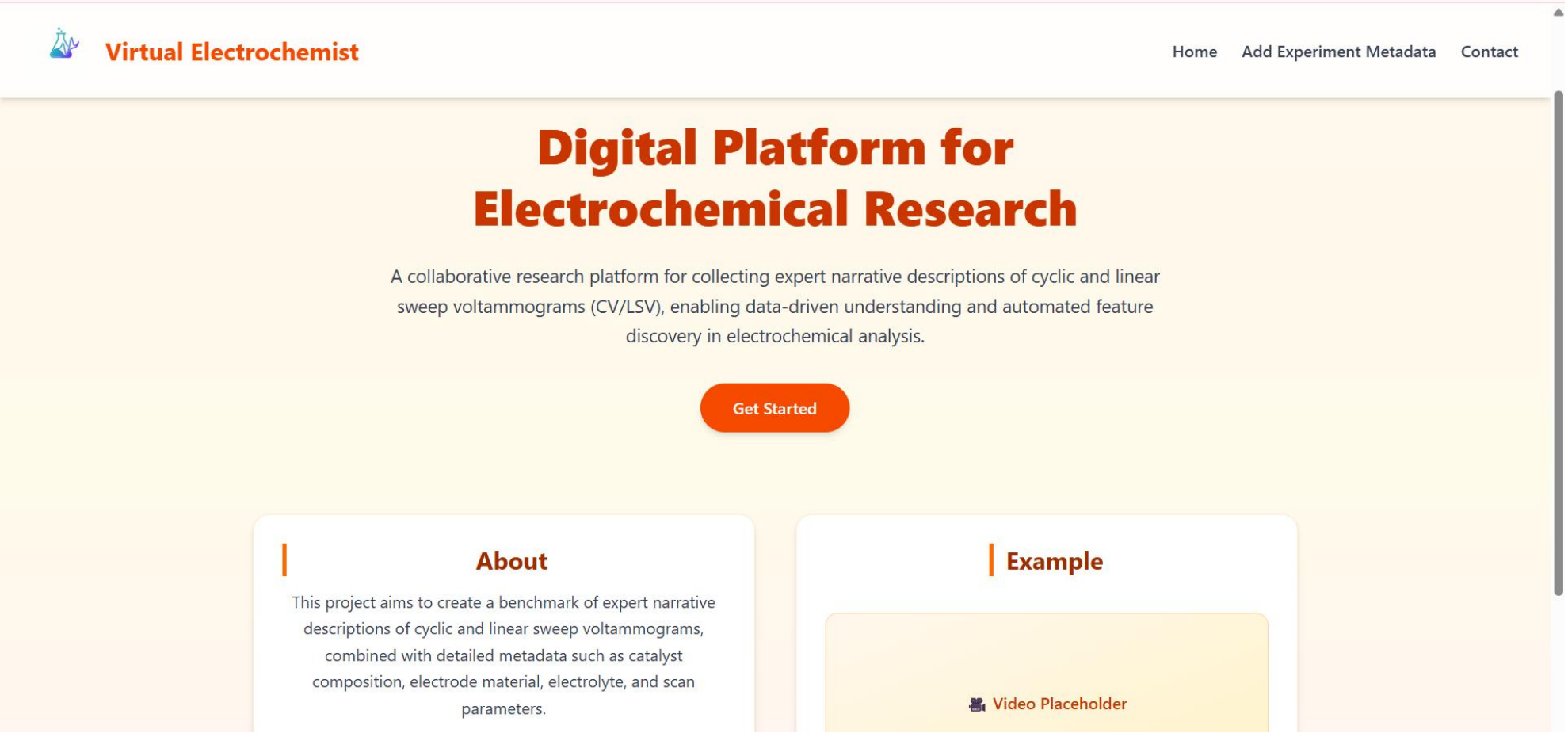
FAIR collection, curation, and management of multidimensional research data

Virtual Electrochemist



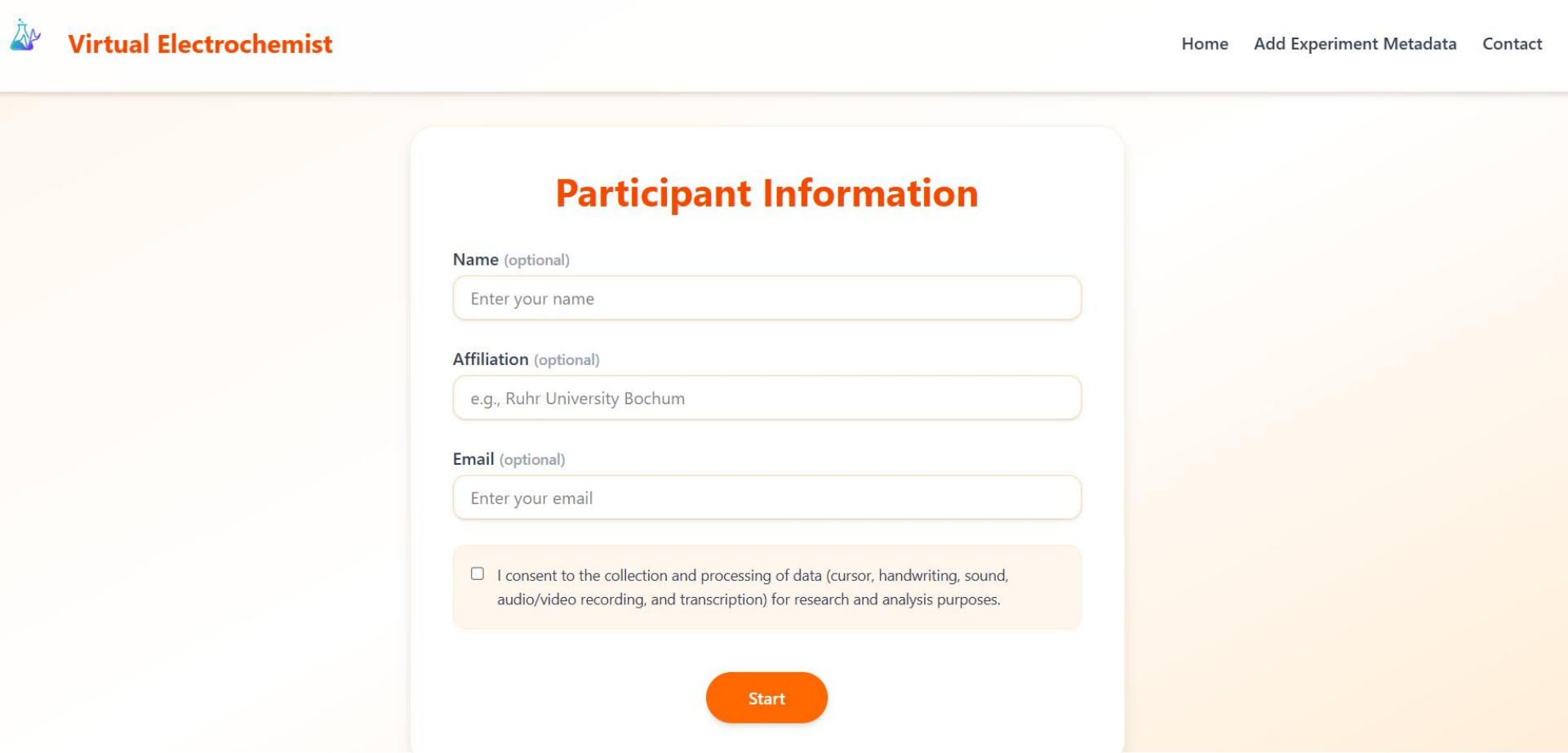
Virtual Electrochemist

Start Page



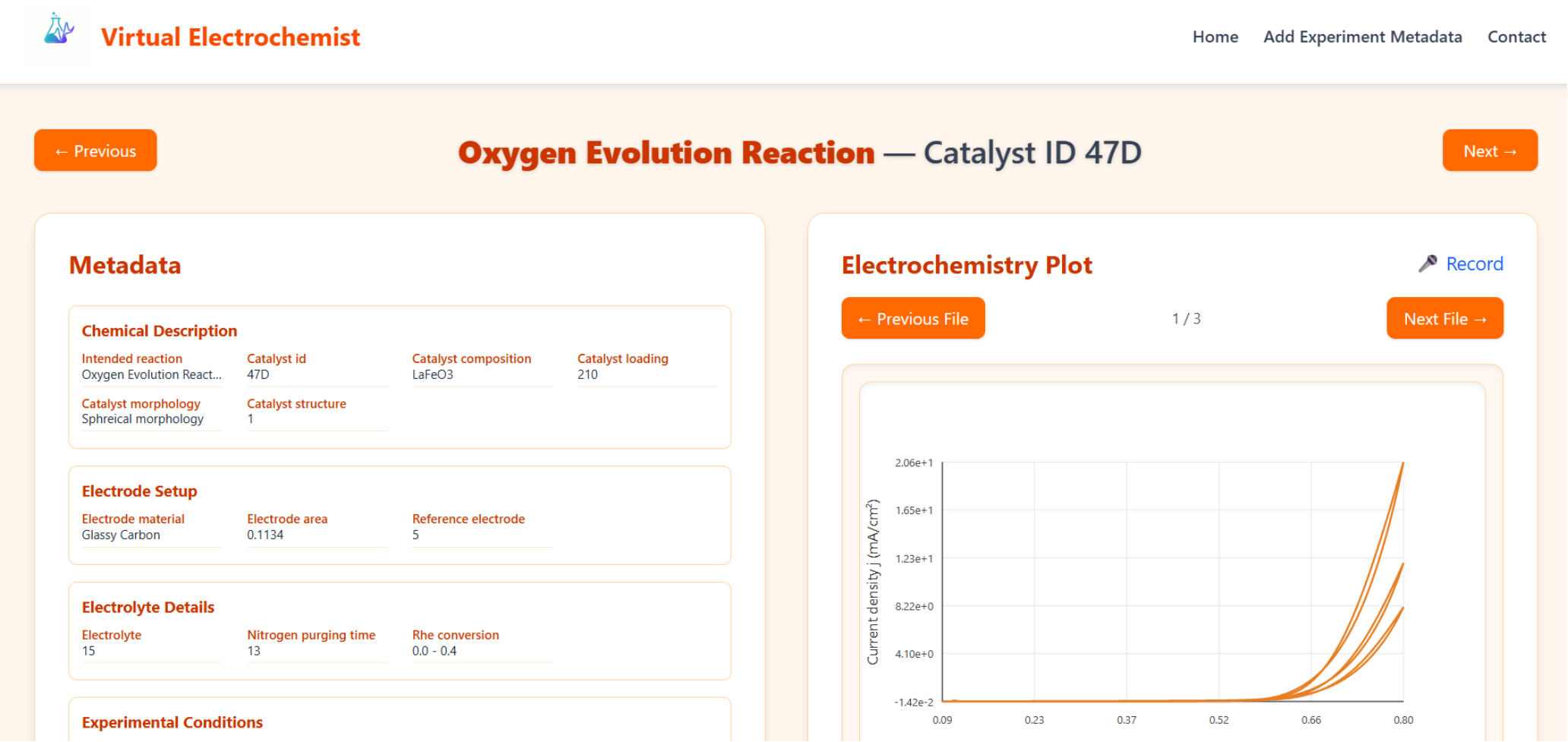
Virtual Electrochemist

Page for receiving the consent from the user that will use the details recorded for postprocessing analysis.



Virtual Electrochemist

Experiment details so the participants give the feedback records.

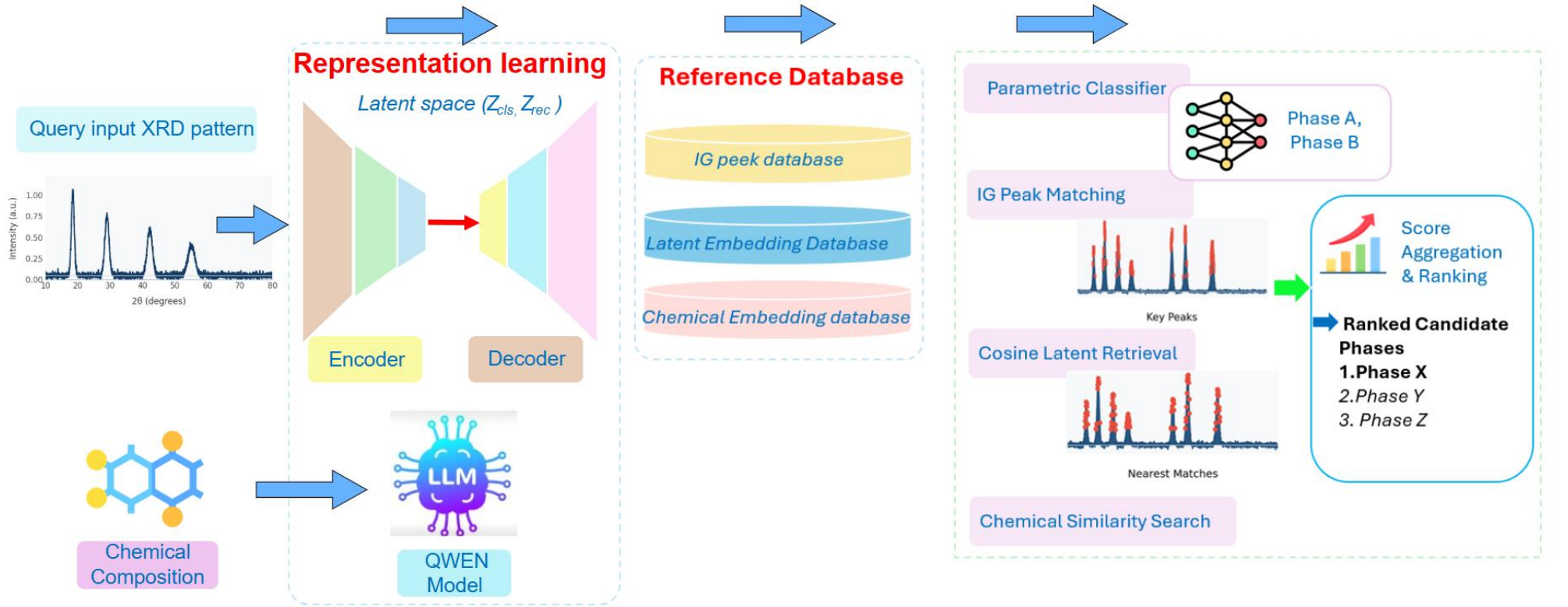


Virtual Electrochemist

Page to add the experimental metadata.

XRD

An **autoencoder** encodes the XRD pattern into a latent representation. If the **chemical composition** is available, a **Qwen model** processes it to generate a complementary embedding.



XRD

A query XRD pattern, together with the chemical composition, is first passed into a **representation learning model** (encoder–decoder architecture, based on a Qwen LLM). The model maps the input into a **latent space embedding** that captures structural and chemical features.

The learned representation is then compared against a **reference database**, which includes:

- IG peak database
- Latent embedding database
- Chemical embedding database

In the final stage, multiple matching strategies are applied, including:
Parametric classification
- IG peak matching
- Cosine similarity retrieval in latent space
- Chemical similarity search

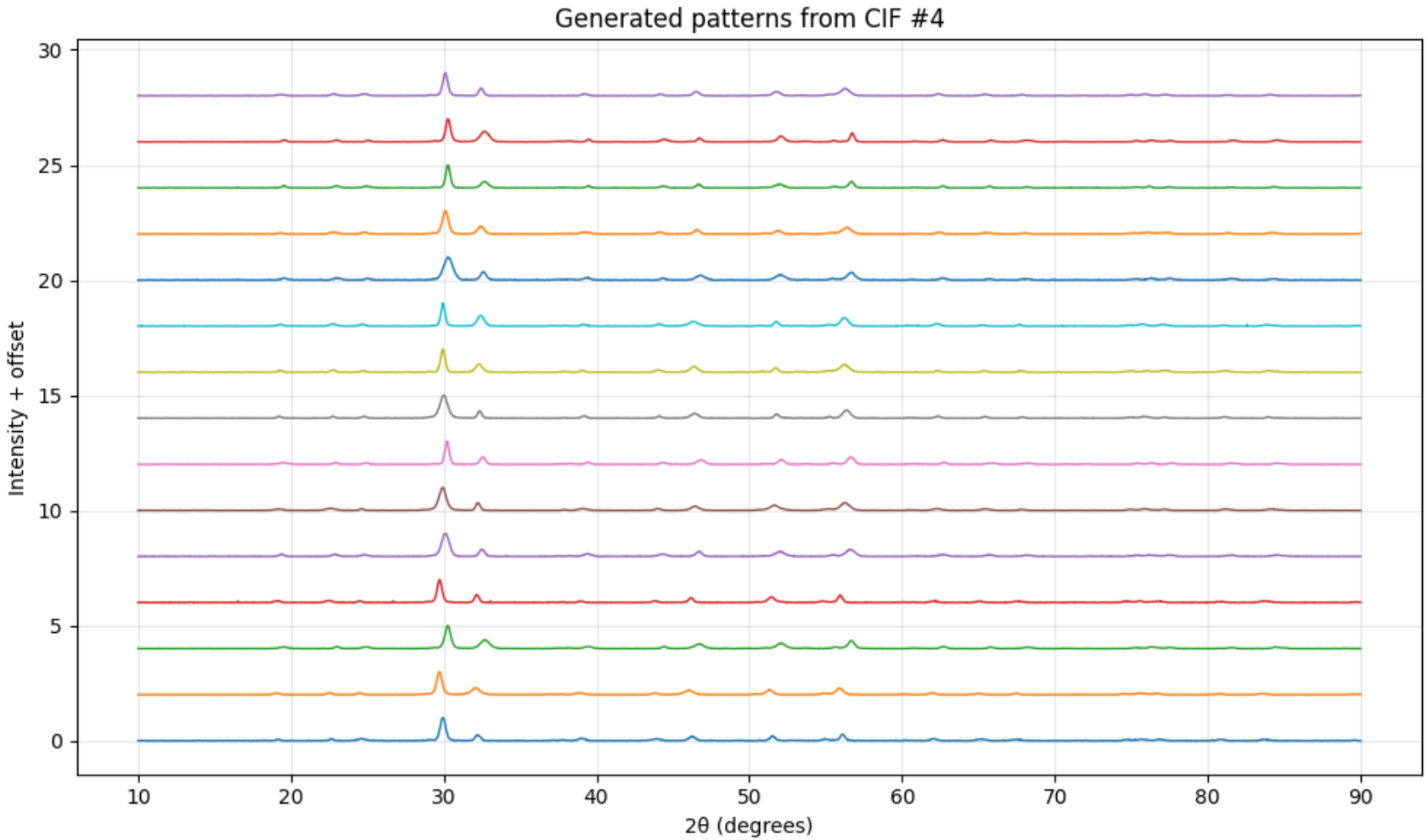
The results are aggregated and ranked, producing a **ranked list of candidate phases** (e.g., Phase X, Phase Y, Phase Z) based on similarity scores.

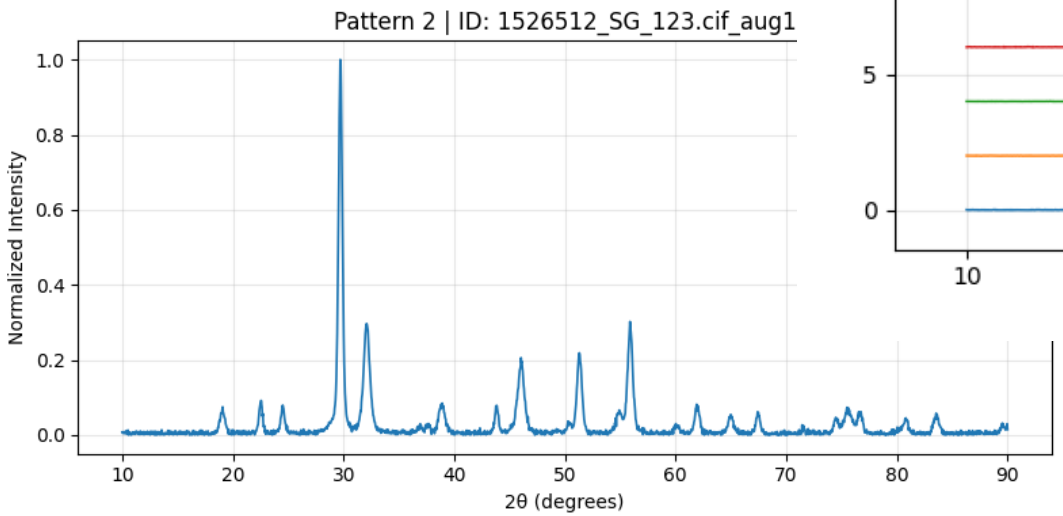
Dataset

The dataset was collected from the Crystallography Open Database (COD) and the Materials Project. In total, we obtained 49,000 CIF files for crystal system classification. To enhance the dataset, we applied data augmentation by generating 15 additional samples from each CIF file.

For space group classification, we used 34,500 CIF files covering 23 space groups. These were carefully selected to construct a balanced dataset suitable for classification tasks.

Augmentation example





Model Performance for Crystal System

The model demonstrates strong performance on the primary dataset, achieving a reconstruction error of 0.000341 on the test set. When evaluated on external datasets, the reconstruction errors remain low (0.0004 for RRUFF and 0.00037 for NiCoAl), suggesting robust generalization across different data distributions.

Test Dataset	Number of CIFS	Model Accuracy	Cosine similarity
10% holdout set from the (MP) and COD dataset	4,900	95.6%	70%
NiCoAl dataset	31	94.4%	62.3%
RRUFF	105	72.0%	66.2%

Model Performance for Space Group

The observed decrease in space group prediction accuracy can be attributed to the inherent difficulty of multi-class classification across 23 distinct space groups. Fine-grained structural distinctions between space groups make this a significantly more challenging task compared to crystal system classification.

Test Dataset	Number of CIFS	Model Accuracy	Cosine similarity
10% holdout set from the (MP) and COD dataset	4,900	88.6%	70%
NiCoAl dataset	31	46.4%	40.8%
RRUFF	105	46%	28%

Composition similarity

Cosine similarity measures the angular similarity between ℓ_2-normalized embedding vectors, capturing compositional proximity independent of magnitude. Higher similarity values indicate chemically related compositions in the learned embedding space.

The examples below show retrieval results using a NiCoAl-based query composition. In both cases, the correct crystal system and space group appear within the top-5 retrieved candidates, demonstrating the effectiveness of embedding-based compositional retrieval.

```
Query: Al0.24Co1.76
True SG: 194
True Crystal System: hexagonal

Top Retrieved Results:

   rank              id   composition   score  space_group crystal_system
0     1  1523557_SG_225  Al0.52Co3.48  1.0000          225          cubic
1     2      mp-1228933         AlCo4  0.9983          166       trigonal
2     3      mp-1188126    Pr2Al2Co15  0.9979          166       trigonal
3     4      mp-1220026      PrErCo17  0.9977          160       trigonal
4     5         mp-2531       Er2Co17  0.9975          194      hexagonal
```

```
Query: Al4Ni3
True SG: 230
True Crystal System: cubic

Top Retrieved Results:

   rank          id   composition   score  space_group crystal_system
0     1    mp-16515        Al4Ni3  1.0000          230          cubic
1     2     mp-1057        Al3Ni2  0.9997          164       trigonal
2     3  mp-1228868          AlNi  0.9981          123     tetragonal
3     4  mp-1192626  Th2(Al3Ni2)5  0.9978           71   orthorhombic
4     5  mp-1228186      Al4Ni3Pt  0.9959          123     tetragonal
```

RUFF Dataset Composition Coverage

This screenshot highlights that most compositions from the RRUFF dataset are not present in the compositional data collected from the Materials Project (MP) and the Crystallography Open Database (COD).

Despite this limited compositional overlap, the model is still able to generalize and perform classification with acceptable accuracy, demonstrating robustness to distribution shift.

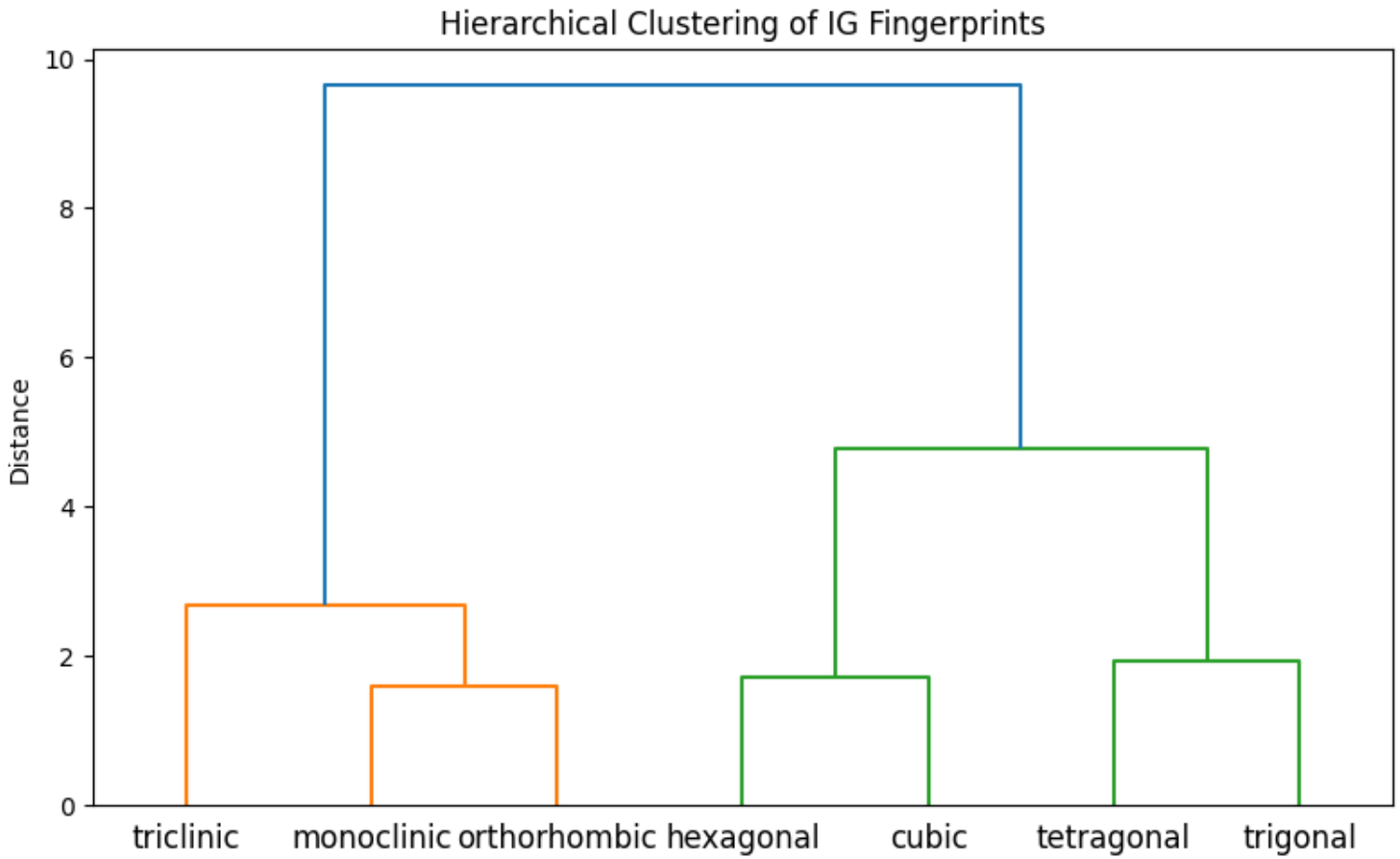
```
Missing compositions:
NaBe4SbO7
Li0.99Al0.99Si2.01O6
MgAlBO4
NaMg3Al6Si6B3O31
Mg3(SiO3)4
K2Ca2Mg(S2O9)2
SrAl3PSO14
KNa3Al4(SiO4)4
Fe1.132Co0.108Ni2.756
Fe(SbS2)2
NaBeSi3O8
BaFe(Si2O5)2
Na6S2ClO8F
CaAl3PSO14
Li0.15Al0.15Si2.85O6
Ca24Al16Si16.008O96
NaH4ClO2
Mg3.68Al4.2Fe0.06B4O16
Ca2Al2FeSi3O13
KAl3(SO7)2
Li0.465Al0.465Si2.535O6
Zn0.04Fe1.24Cu2.12Sn0.2Sb0.04Te0.08Mo0.36As0.12S2.84
Na0.36Ca2.464Mg3.608Ti0.08Al1.368Fe0.84Si7.28O24
Si4H31.99968N8F24
...
Ag3Te2Au
FeSbS
USiO4
Mg2Al3B2HO10
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Hierarchical Clustering of IG Fingerprints

The dendrogram shows clustering of mean Integrated Gradients (IG) attribution patterns for the seven crystal systems.

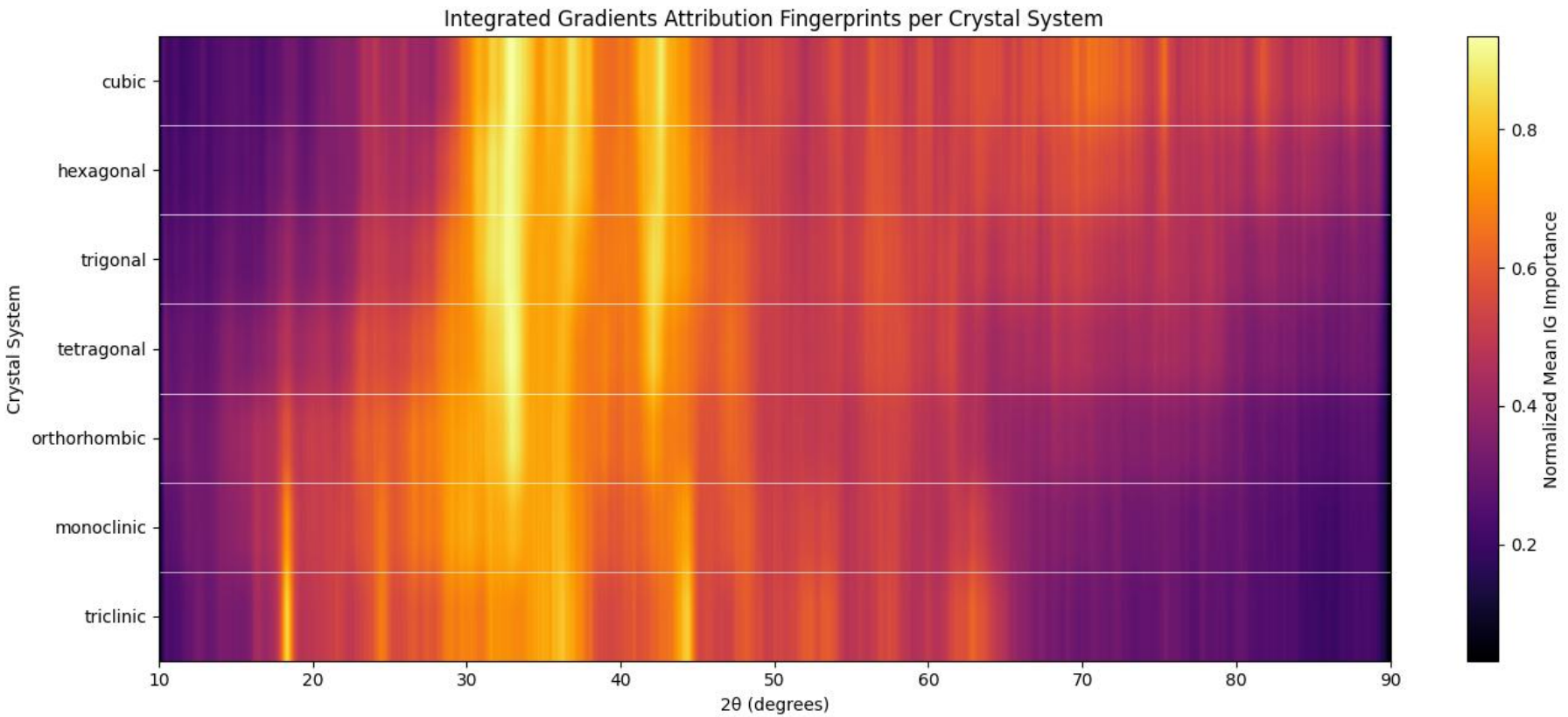
Two main groups emerge: low-symmetry systems (*triclinic, monoclinic, orthorhombic*) cluster together, while higher-symmetry systems (*hexagonal, cubic, tetragonal, trigonal*) form a separate branch.

This indicates that the model relies on distinc diffraction regions to differentiate low- and high-symmetry crystal systems, reflecting meaningful structural relationships in its learned representations.



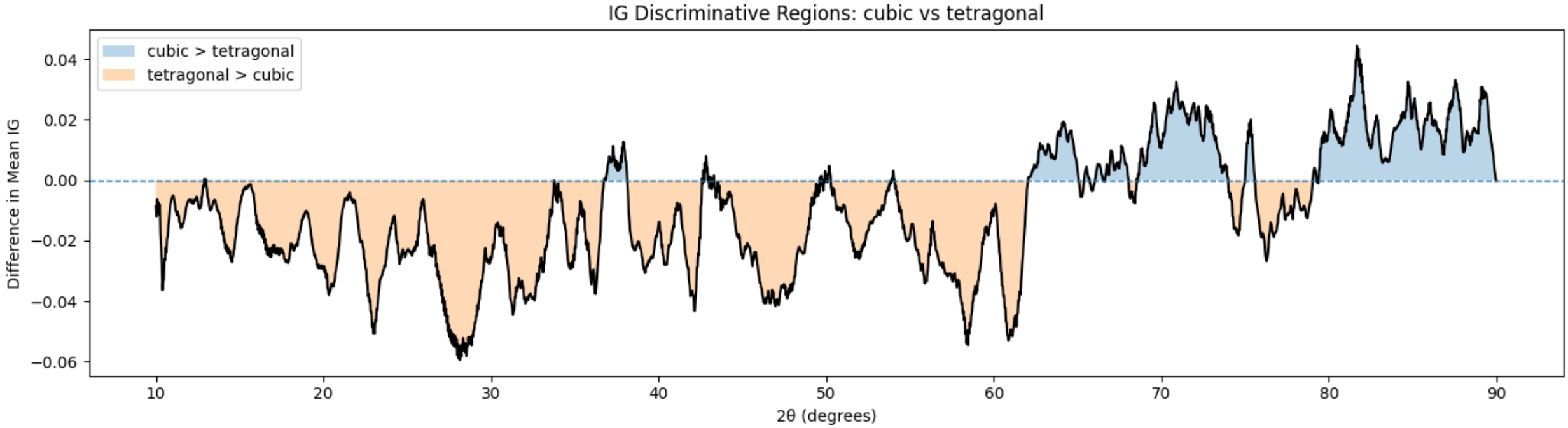
Integrated Gradients Attribution Fingerprints per Crystal System

The heatmap displays the normalized mean IG importance across 2θ for each crystal system. Bright regions mark diffraction angles most influential in the model's decisions, revealing symmetry-dependent attribution patterns.



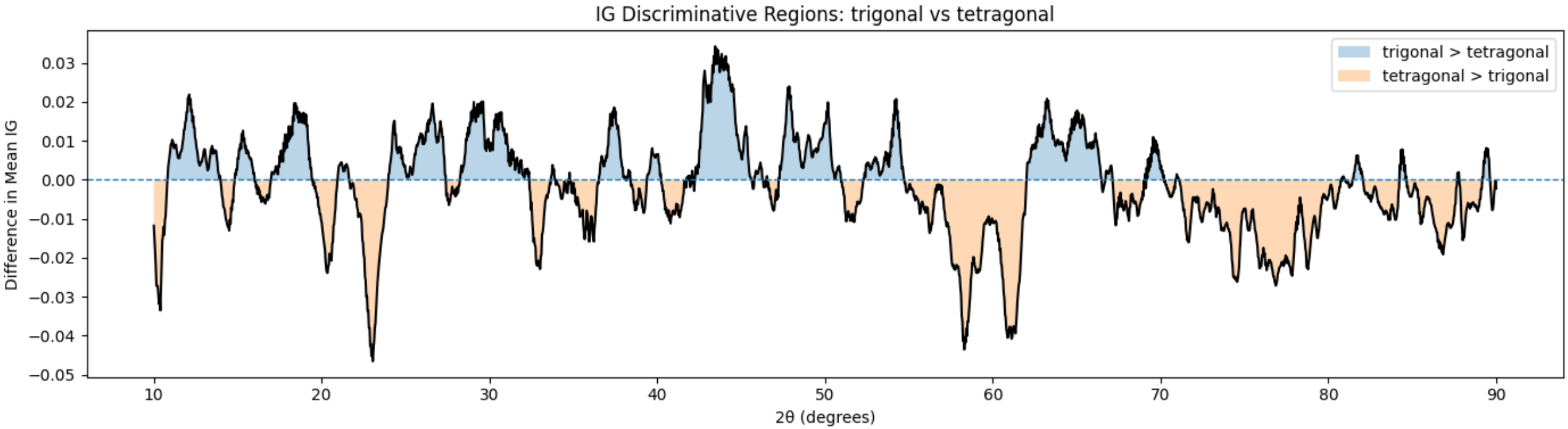
IG Discriminative Regions: Cubic vs. Tetragonal

The plot shows the difference in mean IG importance between cubic and tetragonal systems across 2θ. Positive regions indicate angles more important for cubic classification, while negative regions favor tetragonal. These localized differences reveal the diffraction regions driving the model's distinction between the two symmetries.

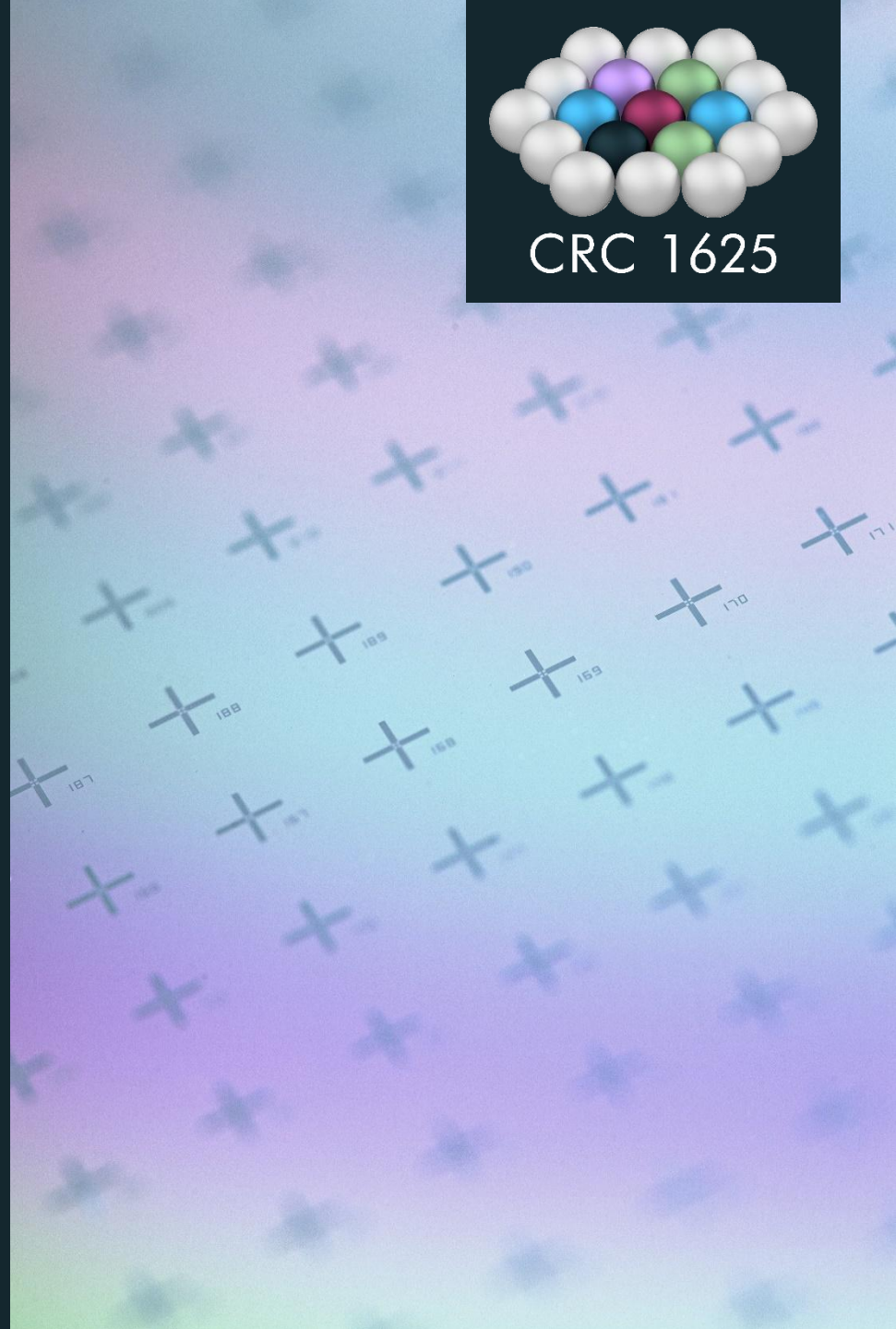


IG Discriminative Regions: Trigonal vs. Tetragonal

The plot shows the difference in mean IG importance between trigonal and tetragonal crystal systems across 2θ. Positive regions indicate diffraction angles more influential for trigonal classification, while negative regions favor tetragonal. The alternating localized peaks highlight the specific diffraction regions the model uses to distinguish between these two symmetry classes..



Feedback and Questions



CRC 1625

Contact
E-Mail: Doaa.Mohamed@ruhr-uni-bochum.de